International Trade and Food Shocks

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Abstract

Does trade make food systems more fragile or more resilient to shocks? We answer this question using a multi-country, multi-sector quantitative trade model covering agriculture, food processing, and the rest of the economy for 23 countries and 29 sectors over 1995–2020. We recover the productivity, preference, land-quality, and policy shocks that rationalize observed changes in trade, output, and expenditure, and use them to assess how access to open markets shapes food-system resilience. We find that openness substantially buffered food-system shocks. Relative to autarky, the annual real consumption growth rate was higher in the open economy by a cross-country median of 3% per year, with much larger gains for small open economies in Latin America, Africa, and Oceania (exceeding 10–20% per year). Six large economies—China, Russia, the EU, Japan, Norway, and Switzerland—show negative average gains, as the specific productivity shocks recovered for these countries would have been better absorbed domestically under autarky. These results suggest that agricultural trade has been a source of resilience, not vulnerability, during the past quarter century.

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1 Introduction

Agricultural trade has returned to the center of policy debates on food security. After decades of declining trade barriers and deeper integration in global food markets, governments are once again turning to tariffs, safeguards, export restrictions, and domestic support to protect farmers and insulate consumers from external shocks. Recent episodes illustrate this shift. China has tightened restrictions on beef imports to protect its cattle sector, the European Union has expanded agricultural safeguards in trade negotiations, Ukraine has introduced minimum export prices on grains and oilseeds, and the United States has raised tariffs across a wide range of imports (Reuters, 2025a,b, 2024, The Guardian, 2025). These policies have emerged against the backdrop of repeated disruptions to global food markets, including the COVID-19 pandemic, the Russian invasion of Ukraine, and climate-related harvest failures. Together, these developments have revived a fundamental question in international trade and food policy: does protectionism make countries more or less food secure?

The case for protection is intuitive. Countries that rely on foreign suppliers may be exposed to world price volatility, export restrictions, and geopolitical risks. Domestic production, supported by tariffs or subsidies, may therefore appear to provide insurance against external shocks. Yet this argument is incomplete. Agricultural markets are linked across countries not only through trade in final food products, but also through intermediate inputs, processed food, and competing uses of land and labor. A tariff that raises the domestic price of one crop may also raise input costs, distort production decisions, and divert resources away from sectors in which the country has comparative advantage. Whether protection buffers or amplifies food-system shocks is therefore a quantitative general-equilibrium question.

In this paper, we provide a first quantitative assessment of this question. We build a multi-country, multi-sector general-equilibrium model of agriculture, food processing, and the rest of the economy. Countries trade through bilateral iceberg trade costs and ad-valorem tariffs, while governments can also support domestic production through output subsidies. Agricultural production features heterogeneous land productivity, which determines both the allocation of land across crops and the quality of land brought into production, following Costinot et al. (2016) and Sotelo (2020). The model is calibrated to EXIOBASE 3 bilateral input-output tables for 23 countries and 29 sectors over 1995–2020, combined with applied bilateral tariffs from the Global Tariff Database (Teti, 2024) and producer-support estimates from the OECD (2025).

Our analysis proceeds in two steps. First, we use the model's exact hat algebra to recover the year-on-year structural shocks that rationalize the observed evolution of trade shares, output, and final expenditure between 1995 and 2020. These shocks include productivity changes in primary agriculture and processed food, country-wide land-quality shocks, preference shifts, and changes in labor and land endowments. This procedure delivers a country-sector-year shock path that captures both gradual changes in global food markets and major disruptions such as the Asian financial crisis, the Global Financial Crisis, and the COVID-19 shock.

Second, we use the recovered shock path to ask how the same shocks would have affected countries in the absence of international trade. Our benchmark counterfactual is autarky, implemented by sending bilateral trade costs to infinity. This counterfactual is deliberately stark, but it provides a transparent benchmark for evaluating whether access to foreign markets made countries more or less resilient to the shocks observed over the past quarter century.

The main result is that trade substantially increased resilience. Under the recovered shocks, the open economy delivered higher annual real consumption growth than autarky by a cross-country median of 3% per year across all 575 country–year pairs; the expenditure-weighted annual average is 3.9%. The average effect masks substantial heterogeneity: small open economies and major agricultural exporters, including Brazil (+11%/yr), Australia (+12%/yr), Canada (+20%/yr), and the Rest-of-Latin-America composite (+10%/yr), gain the most from openness, while large domestically-integrated economies — China (−12%/yr), Russia (−4%/yr), and the EU (−3%/yr) — show negative average gains. Trade acts as a buffer in both directions. When domestic productivity rises, countries can export the surplus rather than absorb a domestic price collapse. When domestic productivity falls, they can substitute toward foreign supply rather than bear the full adjustment at home. In this sense, the counterfactual suggests that protectionism would have amplified, rather than absorbed, the food-system shocks observed between 1995 and 2020.

Our paper contributes to the literature that builds on quantitative models of agriculture and trade. Costinot et al. (2016) show that the evolving geography of comparative advantage is central for quantifying the effect of climate change on agricultural markets, and that trade can dampen welfare losses by allowing countries to adjust production and imports. Sotelo (2020) develops a framework that links land heterogeneity, domestic trade frictions, and agricultural market access. Farrokhi and Pellegrina (2023) shows that trade in agricultural inputs has been central to the rise of modern agriculture. Allen and Atkin (2022) embed a portfolio-choice problem into a many-location, many-crop Ricardian trade model, showing that falling trade costs reshape not only the level but also the volatility of farmer real income. Related quantitative treatments emphasize the role of agriculture in cross-country productivity and welfare differences (Tombe, 2015) and the scope for sectoral reallocation to mediate climate adaptation (Nath, 2025). We build on this literature by incorporating trade policy and producer-support wedges into a multi-sector input-output model of food systems.

Our paper also speaks to the literature on the relationship between trade openness, protectionism, and risk. A long-standing view holds that openness raises income volatility by deepening specialization and exposure to sector-specific shocks (di Giovanni and Levchenko, 2009). Caselli et al. (2020) qualify this view, showing that when country-wide shocks are important, openness can instead *lower* volatility by diversifying the sources of supply and demand across countries; Allen and Atkin (2022) make a related point in an agricultural setting, where the second-moment effects of trade operate through the correlation between local prices and productivity shocks. The classic theoretical counterpart to these quantitative results is Bhagwati and Srinivasan (1976), who characterize optimal trade policy and compensation when openness exposes a country to the risk of market disruption. More recently, Grossman et al. (2023) study endogenous investment in supply-chain resilience in general equilibrium, deriving the policy instruments that decentralize the first best. Our autarky benchmark and subsequent protection counterfactuals connect directly to this risk-versus-gains-from-trade trade-off, but in a setting where the relevant shocks are food-system shocks transmitted through input-output linkages.

Finally, the paper relates to the economics of food security and the national-security rationale for protection. Anderson et al. (2013) survey the political economy of agricultural and food-market distortions, providing the empirical backdrop for the producer-support and trade-policy wedges we construct. A recurring theme in this literature is that uncoordinated, nationally optimal insulation against price shocks can be collectively self-defeating: Martin and Anderson (2011) show that export restrictions and price insulation during commodity price booms amplify world price volatility, and Anderson et al. (2015) quantify the resulting poverty incidence. Bellemare (2015) links food price levels to social unrest, providing the bridge from food-market shocks to broader security concerns. On the theory side, Thompson (1979) provides the canonical economic foundation for the national-defense argument, showing that the first-best response to wartime distortions is typically a targeted domestic subsidy rather than an import tariff, a targeting logic that maps directly onto our decomposition of policy into trade-cost and production-subsidy wedges.

Relative to this literature, the paper makes three contributions. First, it provides a general-equilibrium framework that combines heterogeneous agricultural land allocation, input-output linkages across primary agriculture and food processing, and explicit trade and production-policy wedges. Second, it develops an exact-hat-algebra procedure to recover the structural shocks that rationalize the evolution of global food markets between 1995 and 2020. Third, it uses the recovered shock path to evaluate a first counterfactual benchmark: how the same shocks would have affected countries under autarky. This benchmark provides a foundation for future policy experiments involving unilateral protection, export restrictions, and tariff-war scenarios.

The remainder of the paper is organized as follows. Section 2 develops the theoretical model. Section 4 describes the data and calibration. Section 5 explains the shock-recovery procedure and the counterfactual methodology. Section 6 presents the recovered shocks and the autarky results. Section 7 concludes.

2 Model

Time is discrete, $t = 0, 1, 2, \ldots$. The world has N countries and S sectors partitioned into primary agriculture S_{ag}, processed food S_{food}, and a residual block S_{other}. Each country is endowed with labor $L_{i,t}$ and an exogenously given cultivated cropland area $X_{i,t}$. Within each period the production and trade equilibrium is static; dynamics arise from (i) the gradual reallocation of cropland across crops via a partial-adjustment rule and (ii) the planting-harvest lag, under which current ag output is determined by land committed in the previous period.

2.1 Technology and land allocation

Primary agriculture. Let $T^s_{i,t}$ be the area committed to crop $s \in S_{ag}$. It is *predetermined*: chosen at the end of $t-1$ and fixed within t. Output is Cobb–Douglas in land (fixed), labor, and an intermediate input from each sector:

$$q^s_{i,t} = A^{ag,s}_{i,t} \, (\zeta^s_{i,t} T^s_{i,t})^{\gamma_i^{T,s}} \big(L^s_{i,t}/\gamma_i^{L,s}\big)^{\gamma_i^{L,s}} \prod_{r=1}^{S} \big(m^{rs}_{i,t}/\gamma_i^{M,rs}\big)^{\gamma_i^{M,rs}}, \qquad s \in S_{ag}, \tag{1}$$

with $\gamma_i^{T,s} + \gamma_i^{L,s} + \sum_r \gamma_i^{M,rs} = 1$. Here $A^{ag,s}_{i,t}$ is sectoral TFP and $\zeta^s_{i,t}$ is a land-quality shock (weather, soil health, and the Fréchet selection margin defined below). Since $T^s_{i,t}$ is fixed within the period, the production function has decreasing returns in flexible inputs and yields an upward-sloping supply curve in which the land rent $R^s_{i,t}$ accrues residually to landowners.

Fréchet land allocation. Each plot ω in i's cultivated area draws crop productivities $\{Z_i^s(\omega)\}_{s\in S_{ag}}$ independently from a Fréchet with shape $\varphi > 1$ and country-crop scale a_i^s, $\Pr[Z_i^s(\omega) \leq z] = \exp(-\phi(z/a_i^s)^{-\varphi})$. Each plot is allocated to the highest-return crop, yielding closed-form land shares

$$\Pi_{i,t}^s = \frac{(a_i^s R_{i,t}^s)^\varphi}{\sum_{r\in S_{ag}} (a_i^r R_{i,t}^r)^\varphi}, \qquad s \in S_{ag}, \tag{2}$$

with $\sum_{s\in S_{ag}} \Pi_{i,t}^s = 1$. Higher rent raises a crop's share.

Partial-adjustment land dynamic. Let $\ell_{i,t}^s = T_{i,t}^s / X_{i,t}$. Physical reallocation is costly, so shares evolve as

$$\ell_{i,t+1}^s = (1-\lambda_T)\ell_{i,t}^s + \lambda_T \Pi_{i,t}^s, \qquad T_{i,t+1}^s = \ell_{i,t+1}^s X_{i,t+1}, \qquad \lambda_T \in (0,1]. \tag{3}$$

Processed food and other sectors. For $s \in S_{food} \cup S_{other}$ production is CRS Cobb–Douglas in labor and intermediates only; food additionally uses a CES composite of primary-ag inputs:

$$q_{i,t}^s = A_{i,t}^{food,s} \left(L_{i,t}^s/\gamma_i^{L,s}\right)^{\gamma_i^{L,s}} \left(M_{i,t}^{AG,s}/\gamma_i^{A,s}\right)^{\gamma_i^{A,s}} \prod_{r\in S\setminus S_{ag}} \left(m_{i,t}^{rs}/\gamma_i^{M,rs}\right)^{\gamma_i^{M,rs}}, \qquad s \in S_{food}, \tag{4}$$

$$q_{i,t}^s = A_{i,t}^{oth,s} \left(L_{i,t}^s/\gamma_i^{L,s}\right)^{\gamma_i^{L,s}} \prod_{r=1}^{S} \left(m_{i,t}^{rs}/\gamma_i^{M,rs}\right)^{\gamma_i^{M,rs}}, \qquad s \in S_{other}, \tag{5}$$

with $\gamma_i^{L,s} + \gamma_i^{A,s} + \sum_{r\notin S_{ag}} \gamma_i^{M,rs} = 1$ for food and $\gamma_i^{L,s} + \sum_r \gamma_i^{M,rs} = 1$ for other. The agricultural composite $M_{i,t}^{AG,s}$ is CES across primary-ag inputs with elasticity σ_A. The composite intermediate from any sector r is a CES Armington bundle of varieties sourced from each country j with elasticity σ_{rs}.

2.2 Trade, prices, and policy

Iceberg costs and policy wedges. The purchaser price faced in destination j for a good produced in i in sector s is

$$p^s_{ij,t} = p^s_{i,t}\,\theta^s_{ij,t}\,\frac{1+t^s_{ij,t}}{1+\nu^s_{j,t}}, \tag{6}$$

where $\theta^s_{ij,t} \geq 1$ is the iceberg cost (with $\theta^s_{ii,t} = 1$), $t^s_{ij,t}$ is the bilateral ad-valorem tariff, and $\nu^s_{j,t}$ is the destination-side ad-valorem producer-support wedge.

Final-consumption and intermediate price indices. Households allocate Cobb–Douglas budget shares $\psi^s_{j,t}$ across sectors and CES-Armington across origins within each sector:

$$P^{C,s}_{j,t} = \left[\sum_{i=1}^{N} (p^s_{ij,t})^{1-\sigma_s}\right]^{1/(1-\sigma_s)}, \qquad P^{M,sr}_{j,t} = \left[\sum_{i=1}^{N} \omega^{sr}_i\,(p^s_{ij,t})^{1-\sigma_{sr}}\right]^{1/(1-\sigma_{sr})}. \tag{7}$$

The agricultural composite price in food sector $s \in S_{food}$, country j, is $P^{AG,s}_{j,t} = [\sum_{r\in S_{ag}} \varsigma^s_r (P^{M,rs}_{j,t})^{1-\sigma_A}]^{1/(1-\sigma_A}$

Producer pricing. For $s \in S_{food} \cup S_{other}$, prices follow from CRS Cobb–Douglas zero profit:

$$p^s_{i,t} = (A^{food,s}_{i,t})^{-1}\, w_{i,t}^{\gamma_i^{L,s}}\, (P^{AG,s}_{i,t})^{\gamma_i^{A,s}} \prod_{r\in S\setminus S_{ag}} (P^{M,rs}_{i,t})^{\gamma_i^{M,rs}}, \qquad s \in S_{food}, \tag{8}$$

$$p^s_{i,t} = (A^{oth,s}_{i,t})^{-1}\, w_{i,t}^{\gamma_i^{L,s}} \prod_{r=1}^{S} (P^{M,rs}_{i,t})^{\gamma_i^{M,rs}}, \qquad s \in S_{other}. \tag{9}$$

For $s \in S_{ag}$, the supply schedule is the inverse of (1) given fixed $T^s_{i,t}$:

$$Y^s_{i,t} = \zeta^s_{i,t} T^s_{i,t}\left[A^{ag,s}_{i,t}\, p^s_{i,t}\, w_{i,t}^{-\gamma_i^{L,s}} \prod_r (P^{M,rs}_{i,t})^{-\gamma_i^{M,rs}}\right]^{1/\gamma_i^{T,s}}, \qquad R^s_{i,t} = \gamma_i^{T,s}\,\frac{Y^s_{i,t}}{T^s_{i,t}}. \tag{10}$$

Trade shares and goods clearing. The CES first-order condition gives $\pi^{F,s}_{ij,t} = (p^s_{ij,t}/P^{C,s}_{j,t})^{1-\sigma_s}$ and analogously $\pi^{M,sr}_{ij,t}$. The goods-market clearing identity translates each purchaser-side flow into producer revenue through the policy wedge $W^s_{ij,t} \equiv (1+\nu^s_{j,t})/(1+t^s_{ij,t})$:

$$Y^s_{i,t} = \sum_j W^s_{ij,t}\,\pi^{F,s}_{ij,t}\,X^{F,s}_{j,t} + \sum_{j,r} W^s_{ij,t}\,\pi^{M,sr}_{ij,t}\,X^{M,sr}_{j,t}, \quad X^{M,sr}_{j,t} = \gamma^{M,sr}_j\,Y^r_{j,t}. \tag{11}$$

For $s \in S_{food} \cup S_{other}$ prices are pinned by zero profit so (11) determines quantities $Y^s_{i,t}$; for $s \in S_{ag}$ quantities are pinned by (10) so (11) instead pins the ag producer price $p^s_{i,t}$.

Closure. Final expenditure is allocated $X^{F,s}_{j,t} = \psi^s_{j,t} X^F_{j,t}$ with $\sum_s \psi^s_{j,t} = 1$. The household budget reads

$$X^F_{j,t} = w_{j,t}L_{j,t} + \sum_{s\in S_{ag}} R^s_{j,t}T^s_{j,t} + TR_{j,t} + D_{j,t}, \tag{12}$$

where $D_{j,t}$ is the country's trade deficit (positive = net importer) and $TR_{j,t}$ is tariff revenue net of consumption-side subsidy outlays, rebated lump-sum:

$$TR_{j,t} = \sum_{i,s} \tau^s_{ij,t}\,(\pi^{F,s}_{ij,t}X^{F,s}_{j,t} + \textstyle\sum_r \pi^{M,sr}_{ij,t}\,X^{M,sr}_{j,t}) - \textstyle\sum_s \tilde{\nu}^s_{j,t}\,(X^{F,s}_{j,t} + \textstyle\sum_r X^{M,sr}_{j,t}), \tag{13}$$

with effective rates $\tau^s_{ij,t} \equiv t^s_{ij,t}/(1+t^s_{ij,t})$ and $\tilde{\nu}^s_{j,t} \equiv \nu^s_{j,t}/(1+\nu^s_{j,t})$. The labor market clears as $w_{i,t}L_{i,t} = \sum_s \gamma^{L,s}_i Y^s_{i,t}$.

2.3 Equilibrium

A *recursive competitive equilibrium* is a sequence of prices $\{w_{i,t}, R^s_{i,t}, p^s_{i,t}, p^s_{ij,t}, P^{C,s}_{j,t}, P^{M,sr}_{j,t}, P^{AG,s}_{i,t}\}$, allocations $\{Y^s_{i,t}, X^{F,s}_{j,t}, X^{M,sr}_{j,t}, L^s_{i,t}\}$, and state-variable path $\{T^s_{i,t}\}$ such that, for all i, j, s, r, t: (i) producers optimize given technologies (1)–(5) and predetermined land; (ii) households spend $X^F_{j,t}$ according to the Cobb–Douglas–CES–Armington nest; (iii) producer prices satisfy (8)–(9) and the ag supply equation (10); (iv) all goods markets clear (11) and the labor market clears; (v) Fréchet shares satisfy (2), land dynamics satisfy (3), and initial conditions $\{T^s_{i,0}\}$ are given.

The complete list of exogenous variables and structural parameters is summarized in Table 1 and the derivations of demand systems, supply schedules, and aggregator prices are collected in the online appendix at the project repository.

Table 1: Exogenous variables and time-invariant parameters.

Exogenous variables	$A^{ag,s}_{i,t}, A^{food,s}_{i,t}, A^{oth,s}_{i,t}$ sectoral TFP $\zeta^s_{i,t}$ land-quality shock; $\psi^s_{j,t}$ expenditure shares $L_{i,t}$ labor; $X_{i,t}$ cropland; $D_{j,t}$ trade deficit $t^s_{ij,t}$ tariff; $\nu^s_{j,t}$ producer support; $\theta^s_{ij,t}$ iceberg
Parameters	a^s_i Fréchet scale; φ Fréchet shape; λ_T land adjustment $\omega^{rs}_j, \varsigma^s_r$ CES weights $\gamma^{T,s}_i, \gamma^{L,s}_i, \gamma^{A,s}_i, \gamma^{M,rs}_i$ cost shares $\sigma_A, \sigma_s, \sigma_{rs}$ substitution elasticities

3 The Model in Changes

Following Dekle et al. (2007), we work in exact hat algebra. For any level variable x_t define the gross change $\hat{x}_{t+1} \equiv x_{t+1}/x_t$.

3.1 Shocks in changes

A period-(t+1) shock vector consists of the hats

$$\hat{\Psi}_{t+1} = \left\{\hat{A}^{ag,s}_i, \hat{A}^{food,s}_i, \hat{A}^{oth,s}_i, \hat{\zeta}^s_i, \hat{\psi}^s_i, \hat{L}_i, \hat{X}_i, \hat{\theta}^s_{ij}\right\}_{t+1} \tag{14}$$

together with the period-(t+1) *levels* of the policy and external schedules $\{t^s_{ij,t+1}, \nu^s_{j,t+1}, D_{j,t+1}\}$.

3.2 Equilibrium equations in changes

Land dynamics. The land-share hat under partial adjustment satisfies $\hat{\ell}^s_{i,t+1} = (1 - \lambda_T) + \lambda_T\, \Pi^s_{i,t}/\ell^s_{i,t}$, and $\hat{T}^s_{i,t+1} = \hat{\ell}^s_{i,t+1}\hat{X}_{i,t+1}$. The desired Fréchet share evolves by $\hat{\Pi}^s_{i,t+1} = (\hat{R}^s_{i,t+1}/\hat{\bar{R}}_{i,t+1})^{\varphi}$ where $\hat{\bar{R}}_{i,t+1} = [\sum_{s\in S_{ag}} \Pi^s_{i,t}(\hat{R}^s_{i,t+1})^{\varphi}]^{1/\varphi}$.

Producer prices and ag supply. For $s \in S_{food}$:

$$\hat{p}^s_{i,t+1} = (\hat{A}^{food,s}_{i,t+1})^{-1} \hat{w}^{\gamma_i^{L,s}}_{i,t+1} (\hat{P}^{AG,s}_{i,t+1})^{\gamma_i^{A,s}} \prod_{r \notin S_{ag}} (\hat{P}^{M,rs}_{i,t+1})^{\gamma_i^{M,rs}}, \tag{15}$$

for $s \in S_{other}$:

$$\hat{p}^s_{i,t+1} = (\hat{A}^{oth,s}_{i,t+1})^{-1} \hat{w}^{\gamma_i^{L,s}}_{i,t+1} \prod_{r=1}^{S} (\hat{P}^{M,rs}_{i,t+1})^{\gamma_i^{M,rs}}, \tag{16}$$

and for $s \in S_{ag}$:

$$\hat{Y}^s_{i,t+1} = \hat{T}^s_{i,t+1} \left[\hat{A}^{ag,s}_{i,t+1} (\hat{\zeta}^s_{i,t+1})^{\gamma_i^{T,s}} \hat{p}^s_{i,t+1} \hat{w}^{-\gamma_i^{L,s}}_{i,t+1} \prod_{r=1}^{S} (\hat{P}^{M,rs}_{i,t+1})^{-\gamma_i^{M,rs}} \right]^{1/\gamma_i^{T,s}}, \tag{17}$$

with land rent $\hat{R}^s_{i,t+1} = \hat{Y}^s_{i,t+1} / \hat{T}^s_{i,t+1}$.

Aggregator prices. The purchaser-price hat is

$$\hat{p}^s_{ij,t+1} = \hat{p}^s_{i,t+1} \, \hat{\theta}^s_{ij,t+1} \, \frac{1 + t^s_{ij,t+1}}{1 + t^s_{ij,t}} \, \frac{1 + \nu^s_{j,t}}{1 + \nu^s_{j,t+1}}, \tag{18}$$

and the consumption- and intermediate-price indices follow the CES dual:

$$\hat{P}^{C,s}_{j,t+1} = \Big[\sum_i \pi^{F,s}_{ij,t} (\hat{p}^s_{ij,t+1})^{1-\sigma_s} \Big]^{1/(1-\sigma_s)}, \; \hat{P}^{M,sr}_{j,t+1} = \Big[\sum_i \pi^{M,sr}_{ij,t} (\hat{p}^s_{ij,t+1})^{1-\sigma_{sr}} \Big]^{1/(1-\sigma_{sr})}. \tag{19}$$

The agricultural composite price hat is $\hat{P}^{AG,s}_{i,t+1} = [\sum_{r \in S_{ag}} \pi^{AG,sr}_{i,t} (\hat{P}^{M,rs}_{i,t+1})^{1-\sigma_A}]^{1/(1-\sigma_A)}$.

Trade shares. The CES first-order conditions give the hat trade shares

$$\hat{\pi}^{F,s}_{ij,t+1} = \big(\hat{p}^s_{ij,t+1} / \hat{P}^{C,s}_{j,t+1} \big)^{1-\sigma_s}, \quad \hat{\pi}^{M,sr}_{ij,t+1} = \big(\hat{p}^s_{ij,t+1} / \hat{P}^{M,sr}_{j,t+1} \big)^{1-\sigma_{sr}}. \tag{20}$$

Goods clearing and budget. The hat version of (11), augmented to make $TR_{j,t+1}$ endogenous, is a linear system in the period-$(t+1)$ levels $u^s_i \equiv \hat{Y}^s_i Y^s_{i,t}$ for non-ag sectors and

$v_j \equiv \hat{X}^F_j X^F_{j,t}$:

$$\begin{bmatrix} I - A_{uu} & -A_{uv} \\ -A_{vu} & D_{vv} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} b_u \\ b_v \end{bmatrix}, \tag{21}$$

whose blocks combine the wedge $W^s_{ij,t+1} = (1 + \nu^s_{j,t+1})/(1 + t^s_{ij,t+1})$, the observed period-$t$ shares, the observed final-expenditure shares $X^{F,s}_{j,t}/X^F_{j,t}$, the cost shares $\gamma^{M,sr}_j$, the ag output hats $\hat{Y}^{ag}_{j,t+1} Y^{ag}_{j,t}$ from (17), and the labor and ag-rent terms $\hat{w}_j \hat{L}_j w_j L_j + \sum_{s \in S_{ag}} \hat{R}^s_j \hat{T}^s_j R^s_j T^s_j + D_{j,t+1}$ on the right-hand side.

Welfare. The change in real final consumption (the welfare hat) for country j is

$$\hat{W}_{j,t+1} = \hat{X}^F_{j,t+1} \Big/ \prod_s (\hat{P}^{C,s}_{j,t+1})^{\beta^s_{j,t}}, \qquad \beta^s_{j,t} = X^{F,s}_{j,t}/X^F_{j,t}, \tag{22}$$

which is the unit-free metric we use to evaluate counterfactuals throughout the paper.

4 Data on Trade, Production, and Policy

We combine three main data sets to measure bilateral trade flows and sectoral production, and the two policy instruments of the model: border tariffs and behind-the-border agricultural producer support. We cover the period 1995–2020 and aggregate the data to the 49 regions of EXIOBASE (44 individually identified countries plus 5 Rest-of-World composites), which we further consolidate into 23 model regions by treating the European Union as a single jurisdiction. Sectors are aggregated to 29 categories that preserve the full disaggregation of primary agriculture (15 `Sag` sectors) and food processing (11 `Sfood` sectors) while collapsing the rest of the economy into three broad blocks (raw materials, manufacturing, services). This section gives a brief discussion of the three main data sets; Appendix A provides a detailed account of cleaning, crosswalks, aggregation, and known limitations, as well as a discussion of supporting data sets.

Multi-Regional Input–Output Tables (EXIOBASE 3). EXIOBASE 3 is a global multi-regional input–output (MRIO) database compiled by Stadler et al. (2018). We use the product-by-product (pxp) version in current basic prices (million EUR), which reports, for each year between 1995 and 2020, an intermediate use matrix, a final demand matrix, and a vector of gross output across 49 regions and 200 products. From these tables we construct the model's gross sectoral output $Y_{i,t}^{s}$, bilateral final-good flows and shares $\pi_{ij,t}^{F,s}$, bilateral intermediate-input flows and shares $\pi_{ij,t}^{M,sr}$, intermediate expenditure $X_{j,t}^{M,sr}$, final expenditure $X_{j,t}^{F,s}$, trade deficits $D_{i,t}$, and the cost shares $\gamma_i^{L,s}$, $\gamma_i^{T,s}$, $\gamma_i^{M,sr}$ that enter the production function. EXIOBASE is unusual among MRIO databases in disaggregating primary agriculture into 15 distinct products and processed food into 11, which is what allows the model to operate at the level of detail needed to study food shocks.

Bilateral Tariffs (Global Tariff Database). We use Feodora Teti's Global Tariff Database (GTD; Teti, 2024), which provides applied bilateral tariff rates at the HS6 product level for 200 importers and their trading partners between 1988 and 2021. Teti (2024) documents that the standard alternative WITS silently interpolates missing preferential rates with MFN rates, generating an upward bias in measured tariffs and a downward bias in estimated trade elasticities. The GTD corrects this by reconstructing the actual applied tariff using the schedule of 149 free trade agreement phase-ins and explicit preferential filings. From the GTD we construct the bilateral tariff wedge $t_{ij,t}^{s}$ in the model.

Agricultural Producer Support (OECD PSE). Behind-the-border agricultural policy is measured using the OECD Producer Support Estimate (PSE; OECD, 2025) database, available through the OECD Data Explorer (`DSD_AGR_POLIND@DF_PSE`). The PSE decomposes producer support into market price support (MPS), output- and input-based payments, payments based on current and historical entitlements, and non-commodity payments. We use this decomposition to construct the destination-side ad-valorem wedge $\nu_{j,t}^{s}$, combining direct output-based payments with the residual market price support not already captured by border tariffs. The construction follows a three-step procedure: tariffs from the GTD enter $t^s_{ij,t}$, output-based PSE payments enter $\nu^s_{j,t}$, and residual MPS above the tariff-implied wedge also enter $\nu^s_{j,t}$. This is designed to avoid double-counting border protection in both instruments.

Elasticities We estimate the elasticities using bilateral trade flows and tariffs as in Fontagné et al. (2022). Table 2 shows the elasticities for each product.

Supporting data and crosswalks. Constructing the panels above requires a layer of supporting infrastructure. BACI bilateral trade flows from CEPII (Gaulier and Zignago, 2010) at the HS6 level provide the trade-value weights used to aggregate GTD tariffs from HS6 to the model's sector classification, and to aggregate ISO3 countries into the five EXIOBASE Rest-of-World composites. Crosswalks are constructed between HS6 and the EXIOBASE pxp classification (using HS chapter 01–24 as the agricultural domain) and between OECD PSE commodities and the EXIOBASE pxp Sag products. Appendix A.4 documents each crosswalk, the unmapped residual it leaves, and the implications for the constructed wedges.

5 Methodology: Shock Recovery and Counterfactuals

Throughout we work with the model in exact hat algebra: a gross-change variable $\hat{x} \equiv x_{t+1}/x_t$ describes how a level x moves between consecutive years. We organise the section in two parts. Subsection 5.1 explains how we use observables in two consecutive years to recover the structural shocks — productivity in primary agriculture, food processing, and other sectors; country-wide land-quality shocks; preference shifts; and policy changes — that rationalise the year-on-year movement of the data. Subsection 5.3 explains how we use the recovered shock path as the input to a counterfactual experiment in which the same shocks are imposed under autarky.

Table 2: Trade elasticities, Fontagné, Guimbard & Orefice (2022) method

Sector	Value	(s.e.)	N
Animal products nec	-0.238**	(0.501)	11,990
Beverages	-1.779***	(0.279)	12,034
Cattle	-5.774[a]		
Cereal grains nec	-5.774[a]		
Crops nec	-5.774[a]		
Dairy products	-6.173***	(0.396)	11,991
Fish products	-3.905***	(0.838)	12,034
Food products nec	-8.228***	(0.562)	12,034
Meat animals nec	-5.774[a]		
Meat products nec	-1.188***	(0.228)	12,034
Oil seeds	-5.774[a]		
Paddy rice	-9.685***	(1.488)	9,117
Pigs	-5.774[a]		
Plant-based fibers	-5.774[a]		
Poultry	-16.279***	(2.744)	10,834
Processed rice	-2.350***	(0.507)	11,990
Products of meat cattle	-5.443***	(0.371)	11,949
Products of meat pigs	-7.286***	(0.822)	11,811
Products of meat poultry	-4.305***	(0.499)	11,824
Raw milk	-5.774[a]		
Sugar	-0.854***	(0.296)	12,012
Sugar cane, sugar beet	-5.774[a]		
Vegetables, fruit, nuts	-3.360***	(0.611)	12,034
Wheat	-0.737***	(0.455)	11,319
Wool, silk-worm cocoons	-5.774[a]		
Manufacturing	-18.091***	(0.788)	13,156
Products of Vegetable oils and fats	-5.774[a]		
Raw materials	-8.263**	(3.618)	12,056
Services	-5.774[a]		

Trade elasticity value = $1 + \sigma$ from a PPML panel gravity regression with exporter×year and importer×year fixed effects, estimated on EXIOBASE bilateral trade and tariffs, 1995–2020. Standard errors are heteroskedasticity-robust. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

[a] No reliable own estimate (missing, wrong-signed, or insignificant at the 5% level); replaced by the cross-sector mean of the valid estimates.

5.1 Recovering shocks from observables

For a given year t, calibration delivers the reference data

$$\{Y^s_{i,t},\ X^{F,s}_{j,t},\ X^{M,sr}_{j,t},\ \pi^{F,s}_{ij,t},\ \pi^{M,sr}_{ij,t},\ w_{i,t},\ L_{i,t},\ R^s_{i,t},\ T^s_{i,t},\ t^s_{ij,t},\ \nu^s_{j,t}\}$$

from EXIOBASE 3, the Global Tariff Database, and the OECD Producer Support Estimates. We rebalance the bilateral final-demand cube so that goods market clearing holds exactly in levels at year t. For each year transition $t \to t+1$ we use the analogous calibration at $t+1$ to back out the structural shocks that, under the model, take the year-t reference to the year-$(t+1)$ reference.

Direct shocks. Several shocks are read directly from the calibrated panels:

- Bilateral tariff changes enter as the period-$(t+1)$ ad-valorem levels $t^s_{ij,t+1}$.
- Destination-side producer-support changes enter as $\nu^s_{j,t+1}$.
- The labor endowment hat $\hat{L}_i = L_{i,t+1}/L_{i,t}$ is read from the cost-share identity that pins $L_{i,t}$ in each year's calibration.
- Total cropland $\hat{X}_i$ is set to one in our baseline (for the moment).

Preference shocks. The preference hat $\hat{\psi}^s_j$ is identified by the residual of the final-demand decomposition:

$$\hat{\psi}^s_j \;=\; \frac{X^{F,s}_{j,t+1}/X^{F,s}_{j,t}}{\hat{X}^F_j}, \qquad \hat{X}^F_j \;=\; \frac{X^F_{j,t+1}}{X^F_{j,t}}. \tag{23}$$

Land-use dynamics. The land-use hat $\hat{T}^s_i$ is *predetermined* by the model's partial-adjustment rule:

$$\Pi^s_{i,t} \;=\; \frac{(a^s_i R^s_{i,t})^{\phi}}{\sum_{r\in S_{ag}}(a^r_i R^r_{i,t})^{\phi}}, \qquad \ell^s_{i,t+1} \;=\; (1-\lambda_T)\,\ell^s_{i,t} + \lambda_T\,\Pi^s_{i,t}, \qquad \hat{T}^s_i \;=\; \frac{\ell^s_{i,t+1}\cdot X_{i,t+1}}{T^s_{i,t}}. \tag{24}$$

Equation (24) is the same dynamic that governs the model when it is solved forward in time, so the recovered $\hat{T}$ is internally consistent: each year-pair's land reallocation responds to the previous period's rents through the Fréchet desired share, smoothed by partial adjustment.

Productivity. After recovering price changes, the structural productivity shocks follow directly from the zero-profit / ag-supply equations of the model:

$$\hat{A}_i^{food,s} = \frac{\hat{w}_i^{\gamma_i^{L,s}} \left(\hat{P}_i^{AG,s}\right)^{\gamma_i^{A,s}} \prod_{r \notin S_{ag}} \left(\hat{P}_i^{M,rs}\right)^{\gamma_i^{M,rs}}}{\hat{p}_i^s}, \qquad s \in S_{food}, \tag{25}$$

$$\hat{A}_i^{oth,s} = \frac{\hat{w}_i^{\gamma_i^{L,s}} \prod_{r=1}^{S} \left(\hat{P}_i^{M,rs}\right)^{\gamma_i^{M,rs}}}{\hat{p}_i^s}, \qquad s \in S_{other}, \tag{26}$$

$$\hat{A}_i^{ag,s} \cdot \hat{\zeta}_i^{s\gamma_i^{T,s}} = \left(\frac{\hat{Y}_i^{obs,s}}{\hat{T}_i^s}\right)^{\gamma_i^{T,s}} \cdot \frac{\hat{w}_i^{\gamma_i^{L,s}} \prod_{r=1}^{S} \left(\hat{P}_i^{M,rs}\right)^{\gamma_i^{M,rs}}}{\hat{p}_i^s}, \qquad s \in S_{ag}. \tag{27}$$

Equations (25)–(26) are point-identified by the price hats and the cost-share parameters. Equation (27) only identifies the *combined* ag shock $\hat{A}_i^{ag,s} \cdot \hat{\zeta}_i^{s\gamma_i^{T,s}}$, because the supply equation enters $\hat{A}^{ag}$ and $\hat{\zeta}$ only through their product. We resolve this in two complementary ways. In the *land-country* decomposition — our baseline — we impose that the geometric mean of the within-country sectoral $\hat{A}_i^{ag,s}$ across primary-ag sectors equals one,

$$\sum_{s \in S_{ag}} \log \hat{A}_i^{ag,s} \; = \; 0 \quad \Longrightarrow \quad \log \hat{\zeta}_i \; = \; \frac{\sum_{s \in S_{ag}} \log\left(\hat{A}_i^{ag,s} \cdot \hat{\zeta}_i^{\gamma_i^{T,s}}\right)}{\sum_{s \in S_{ag}} \gamma_i^{T,s}}, \tag{28}$$

which assigns the country-wide common component of ag productivity to land quality $\hat{\zeta}_i$ and the within-country sectoral variation to TFP $\hat{A}_i^{ag,s}$. In a robustness specification we use the *all-TFP* decomposition, which sets $\hat{\zeta}_i \equiv 1$ and loads all variation onto $\hat{A}_i^{ag,s}$. The two decompositions are observationally equivalent at the level of every model object except the labelling of the ag shock.

What is recovered, what is assumed. The recovery procedure delivers a complete, country-sector-year-specific shock vector ($\hat{A}^{ag}$, $\hat{A}^{food}$, $\hat{A}^{oth}$, $\hat{\zeta}$, $\hat{\psi}$, $\hat{L}$, $\hat{T}$, t_{new}, ν_{new}) for every transition $t \rightarrow t+1$ in the 1995–2020 panel, together with the recovered price system ($\hat{p}$, $\hat{P}^C$, $\hat{P}^M$, $\hat{P}^{AG}$) used to compute it. The preliminary structural assumptions on parameters are summarised in Section 5.2.

5.2 Parameter values

Table 3 lists the parameter values used in the recovery procedure and in the counterfactual that follows. We plan to update most of these values from the data in the next versions of the model

Table 3: Parameter values used in the calibration (preliminary).

Parameter	Value	Role / source
σ_F^s	sector-specific, range [1.25, 7.28], mean 4.0	Final-demand Armington (**?**)
σ_M^{sr}	4 uniform across sector pairs	Intermediate-demand Armington
σ_A	3	Substitution inside the ag composite
φ	5	Fréchet shape parameter
λ_T	0.20	Speed of land partial adjustment
$\gamma_i^{T,s}$	0.20 (primary ag), 0 otherwise	Land cost share, fixed pre-calibration
a_i^s	1 (placeholder)	Fréchet scale; absorbed into $\hat{\zeta}$
$\hat{\theta}_{ij}^s$	1 (no iceberg shock)	Bilateral iceberg-cost change
$\hat{X}_i$	1 (no land-area shock)	Total-cropland change
$\gamma_i^{L,s}$, $\gamma_i^{A,s}$, $\gamma_i^{M,rs}$	from data	EXIOBASE basic-price cost shares
$\pi_{ij,t}^{F,s}$, $\pi_{ij,t}^{M,sr}$, $\pi_{i,t}^{AG,sr}$	from data	EXIOBASE bilateral shares
$t_{ij,t}^s$, $\nu_{j,t}^s$	from data	GTD tariffs and OECD PSE

Preliminary nature of the calibration. We emphasise that the parameter choices reported in Table 3 are deliberately conservative and, where they take a fixed value, motivated by either (a) standard values from the broader trade-elasticity literature ($\sigma = 4$, $\sigma_A = 3$, $\varphi = 5$, $\lambda_T = 0.20$), or (b) values at which the model's hat-algebra equations are numerically well-behaved while remaining within an empirically plausible range. The Armington elasticities for final demand, σ_F^s, are taken from **?**, who estimate product-level trade elasticities from gravity regressions on disaggregated bilateral trade flows using the BACI dataset. We map their estimates to our 29-sector classification; the resulting values range from 1.25 (meat animals) to 7.28 (plant-based fibres) with a cross-sector mean of 4.0. Intermediate-demand Armington elasticities σ_M^{sr} are set to a uniform 4, consistent with the standard value used in the quantitative trade literature (**?**). The land-cost share $\gamma_i^{T,s} = 0.20$ for primary agriculture is the clearest fixed-parameter choice: the supply equation $\hat{Y} = \hat{T} \cdot (\ldots)^{1/\gamma^T}$ has a $1/\gamma^T$ amplification that becomes numerically intractable for the small values (≈ 0.05–0.07) that one would otherwise derive from EXIOBASE factor shares. We choose 0.20 as an empirically plausible value that delivers a workable supply elasticity ($1/\gamma^T = 5$); we plan to relax this in the next iteration by estimating $\gamma_i^{T,s}$ jointly with the productivity hats from observed land-rent data.

More generally, this calibration is a deliberate first pass. In subsequent versions of the paper, we intend to *recover* the parameters of interest — the Armington elasticities σ_F^s and σ_M^{sr}, the Fréchet shape φ, the land-adjustment speed λ_T, and the land cost shares $\gamma_i^{T,s}$ — directly from the data, exploiting the cross-sectional and time-series variation in the EXIOBASE bilateral trade panel, the OECD producer-support panel, and supplementary land-use data from FAOSTAT and MapSPAM.

5.3 Counterfactuals: autarky

A counterfactual asks how welfare and quantities would have evolved under an alternative trade or policy regime, holding fixed the underlying structural shocks. Operationally, this means feeding the same shock vector ($\hat{A}^{ag}$, $\hat{A}^{food}$, $\hat{A}^{oth}$, $\hat{\zeta}$, $\hat{\psi}$, $\hat{L}$, $\hat{T}$) into the model under a counterfactual setting for the policy / trade-cost vector and re-solving for prices and quantities.

The autarky limit. In our headline counterfactual we send iceberg trade costs to infinity for all $i \neq j$, $\hat{\theta}^s_{ij} \to \infty$. This collapses bilateral trade shares to the diagonal:

$$\hat{\pi}^{F,s}_{ij} \to 0 \quad \text{for } i \neq j, \qquad \pi^{F,s}_{jj,t+1} \to 1, \tag{29}$$

and analogously for π^M. The consequence is that every country becomes a closed economy: each sectoral aggregator collapses to the country's own producer price hat,

$$\hat{P}^{C,s}_j \;=\; \hat{p}^s_j, \qquad \hat{P}^{M,sr}_j \;=\; \hat{p}^s_j, \qquad \hat{P}^{AG,s}_j \;=\; \left(\sum_{r \in S_{ag}} \pi^{AG,rs}_{j,t} \left(\hat{p}^r_j\right)^{1-\sigma_A}\right)^{1/(1-\sigma_A)} . \tag{30}$$

Solver. We solve the autarky system by a damped fixed-point iteration: Cobb–Douglas zero profit gives the food and other prices in closed form given a guess of $\hat{p}^{ag}$; the ag supply equation gives $\hat{Y}^{ag}$; the Leontief inverse delivers non-ag output; the ag goods-clearing residual drives an exponential update on $\hat{p}^{ag}$; the income identity updates $\hat{X}^F$. With step damping of 0.10 on prices and 0.40 on final expenditure, this converges for the majority of country–year cells. For numerically stiff cases — primarily Canada, Switzerland, the UK, Indonesia, Korea, Norway, and Taiwan, where the autarky equilibrium under the recovered shocks requires agricultural prices far from their open-economy values — we employ a continuation (homotopy) method: the solution is traced from an analytically computed starting point at $\lambda = 1$ (the open-economy limit) to $\lambda = 0$ (full autarky) in steps, with a supply-adaptive Newton step to update agricultural prices at each stage. The combined cascade achieves convergence to a relative residual below 10^{-5} for all 575 country–year pairs. We discuss the implications for the welfare results in Section 6.

Welfare hat. For each country and each year-pair, we define the welfare hat as the change in real final consumption,

$$\hat{W}_j = \frac{\hat{X}_j^F}{\hat{P}_j}, \qquad \log \hat{P}_j = \sum_s \beta_j^s \log \hat{P}_j^{C,s}, \qquad \beta_j^s = \frac{X_{j,t}^{F,s}}{X_{j,t}^F}, \tag{31}$$

where $\hat{P}_j^{C,s}$ is the consumer price index computed under the relevant scenario. The *welfare cost of autarky* for country j in year-pair $t \to t+1$ is

$$\Delta \log W_{j,t} \equiv \log\big(\hat{W}_{j,t}^{open}\big) - \log\big(\hat{W}_{j,t}^{aut}\big), \tag{32}$$

which is positive when openness yields higher real consumption growth than autarky for that country in that year. We sum (32) along the 25-year shock path to obtain the cumulative welfare cost, and aggregate across countries using the year-t share of world final expenditure as weights to obtain a global figure.

Generalisation. The same machinery applies to any counterfactual that can be expressed as a change in tariffs, subsidies, or iceberg costs at $t+1$. The autarky limit is one extreme; unilateral liberalisation, tariff-war escalations, and targeted export bans are obtained by setting the relevant entries of $t_{ij,t+1}^s$, $\nu_{j,t+1}^s$, and $\hat{\theta}_{ij}^s$ appropriately and re-solving. We use the autarky limit here because it provides the natural benchmark against which any partial protection scheme can be compared.

6 Results

This section reports the two empirical exercises of the paper. Subsection 6.1 presents the structural shock vector recovered from the 1995–2020 panel and validates it against well-known macroeconomic events. Subsection 6.2 reports the autarky counterfactual.

6.1 The recovered shock path, 1995–2020

The exact-hat-algebra procedure of Section 5.1 delivers a complete shock vector for every (country, sector, year) cell. Three features anchor its credibility. First, the recovered labor-endowment hats $\hat{L}_i$ pick up every major macroeconomic shock in the panel exactly where one would expect to see it (Figure 1). Second, the recovered productivity hats are well-distributed across countries and years with reasonable dispersion (Figure 2). Third, the cross-country pattern of cumulative ag productivity — carried by the country-wide land-quality factor $\hat{\zeta}_i$ under the land-country split of (28) — tracks the well-documented re-balancing of global agriculture toward emerging economies over the past quarter century.

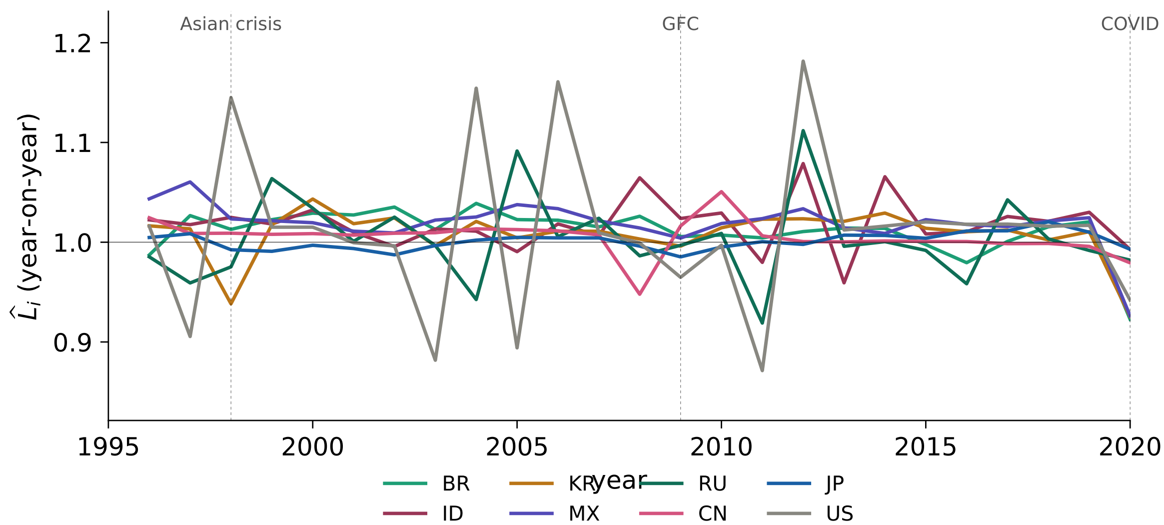

Figure 1: **Labor-endowment hats pick up every major macro shock.** Year-on-year labor-endowment hat $\hat{L}_i$ for selected countries, recovered from the cost-share identity that pins L_i each year. Dashed vertical lines mark the three benchmark events in the panel: the 1997–98 Asian financial crisis ($\hat{L}_{ID} = 0.45$, $\hat{L}_{KR} = 0.68$); the 2008–09 Global Financial Crisis ($\hat{L}_{RU} = 0.78$, $\hat{L}_{MX} = 0.86$, while emerging Asia keeps growing); and the 2019–20 COVID-19 disruption ($\hat{L}_{BR} = 0.75$, $\hat{L}_{MX} = 0.84$, with East Asia largely shielded).

Figure 2 maps the median sector-level productivity shock for each country–year cell, separately for primary agriculture (left panel) and processed food (right panel), expressed as a percentage deviation from no change. Countries are ordered from top to bottom by mean absolute agricultural shock size. Agricultural productivity is considerably more volatile than food processing: shocks range from −31% to +57% with a standard deviation of 14 percentage points, against −45% to +32% and a standard deviation of 9 percentage points for food. The most agriculturally volatile economies are Korea (KR), Australia (AU), the food-exporting rest-of-world aggregate (WF), Japan (JP), and South Africa (ZA), all exhibiting alternating runs of sharp gains and losses. The food-sector heatmap reveals a broad downward trend across most countries over the sample, consistent with ongoing efficiency gains in food processing.

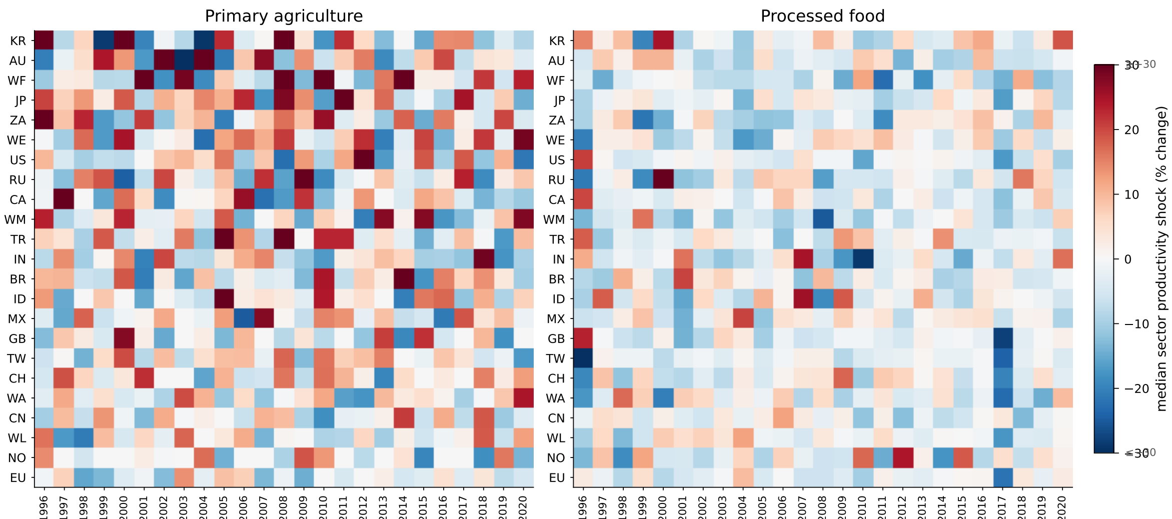

Figure 2: **Agricultural productivity is more volatile than food processing.** Median sector-level productivity shock by country and year, 1996–2020. Each cell shows the median of $(\hat{A}_i^s - 1) \times 100$ across sectors within the relevant category (primary agriculture, left; processed food, right). Red indicates above-trend productivity; blue indicates below-trend. The colour scale is capped at ±30%; a small number of cells exceed this range. Countries are sorted by mean absolute agricultural shock, most volatile at top.

Cumulative productivity, 1995–2020. Under the land-country normalisation, within-country sectoral productivity hats have a geometric-mean cumulative of exactly one by construction; cross-country productivity differences load onto the country-wide land-quality factor $\hat{\zeta}_i$. Cumulative $\hat{\zeta}_i^{25}$ is dominated by China (46×), India (7.5×), Brazil (6.4×), Korea (4.6×), and Mexico (3.3×); developed economies cluster around or below one (US 0.60, EU 0.99, Japan 0.03). The qualitative pattern matches the well-documented re-balancing of global agriculture toward emerging economies; the very large China multiple is partly an artefact of the small land-cost share $\gamma^T = 0.20$ in the supply equation and partly a reflection of the genuinely large structural transformation of Chinese agriculture in the 1990s and 2000s.

Cumulative $\hat{A}^{food}$ shows Korea as the only country with persistent food-TFP gains (1.80×); large economies with mature food industries (US, EU, Japan, Australia) show 30–45% cumulative declines, consistent with sectoral output growth that fell short of input-cost growth over the period. Preference hats $\hat{\psi}_j^s$ display sector-level dispersion that widens in crisis years (1998, 2002, 2008–09, 2020), consistent with consumption rebalancing under disruption.

6.2 The autarky counterfactual

We feed the recovered shock vector into the closed-economy CGE described in Section 5.3, country by country, year by year. For each (j,t) we compute the welfare gain from trade

$$g_{j,t} = \frac{\hat{W}_{j,t}^{open}}{\hat{W}_{j,t}^{aut}} - 1, \tag{33}$$

the percentage by which real consumption growth under the open economy exceeds the autarky counterfactual in that year.

Convergence. The closed-economy solver converges to a relative residual below 10^{-5} for all 575 country–year pairs (23 countries × 25 transitions). Previously intractable cases — Canada, Switzerland, the UK, Indonesia, Korea, Norway, and Taiwan — are now solved by a continuation (homotopy) method with analytic initialisation and a supply-adaptive Newton step. For these economies, the autarky equilibrium under the recovered shocks requires agricultural prices to deviate substantially from their open-economy levels; the standard fixed-point iteration fails to converge from an uninformed starting point, whereas the homotopy approach traces the solution continuously from an analytically computed initial condition.

The global welfare cost of autarky. The expenditure-weighted average annual welfare gain from trade across all 575 country–year pairs is approximately 3.9%, with a cross-country median of 3.0% per year. Summed over all year-pairs and weighted by year-t shares of world final expenditure, the cumulative welfare statistic is

$$\sum_{t}\sum_{j}\omega_{j,t}\,\Delta\log W_{j,t} = 0.96, \tag{34}$$

corresponding to an annual average of 0.038 log points, or 3.9% in level terms. The wide cross-country dispersion — with mean annual gains ranging from −11.7% (China) to +20% (WE, CA) — is itself informative: small open agricultural exporters benefit far more from trade than large internally-heterogeneous economies. The year 2016–17 is an outlier, generating unusually large welfare costs of autarky for commodity exporters (South Africa +197%, Korea +196%, Australia +182% in that single year) owing to the combination of large recovered productivity shocks and high trade dependence; excluding that year, the annual cross-country mean is 3.9% and the expenditure-weighted aggregate is 0.72.

Country-level heterogeneity. Table 4 reports the mean annual welfare gain from trade for all 23 economies in the sample. The ranking is intuitive. The largest gains from openness accrue to small, trade-dependent agricultural exporters: the European rest-of-world composite (WE: +20%/yr), Canada (CA: +20%/yr), the Western African composite (WA: +19%/yr), the food-exporting composite (WF: +16%/yr), Australia (AU: +12%/yr), Brazil (BR: +11%/yr), Mexico (MX: +11%/yr), and the Latin-American composite (WL: +10%/yr). Canada's large gain reflects extreme recovered agricultural productivity shocks — annual swings of −22% to +48% in median crop-sector productivity — that make the autarky price adjustment particularly severe. The United Kingdom (+8%/yr), Indonesia (+5%/yr), Korea (+5%/yr), Turkey (+5%/yr), and India (+5%/yr) are in an intermediate group.

A second group — Taiwan (+0.7%/yr), the Middle-East composite (WM: +0.8%/yr) —

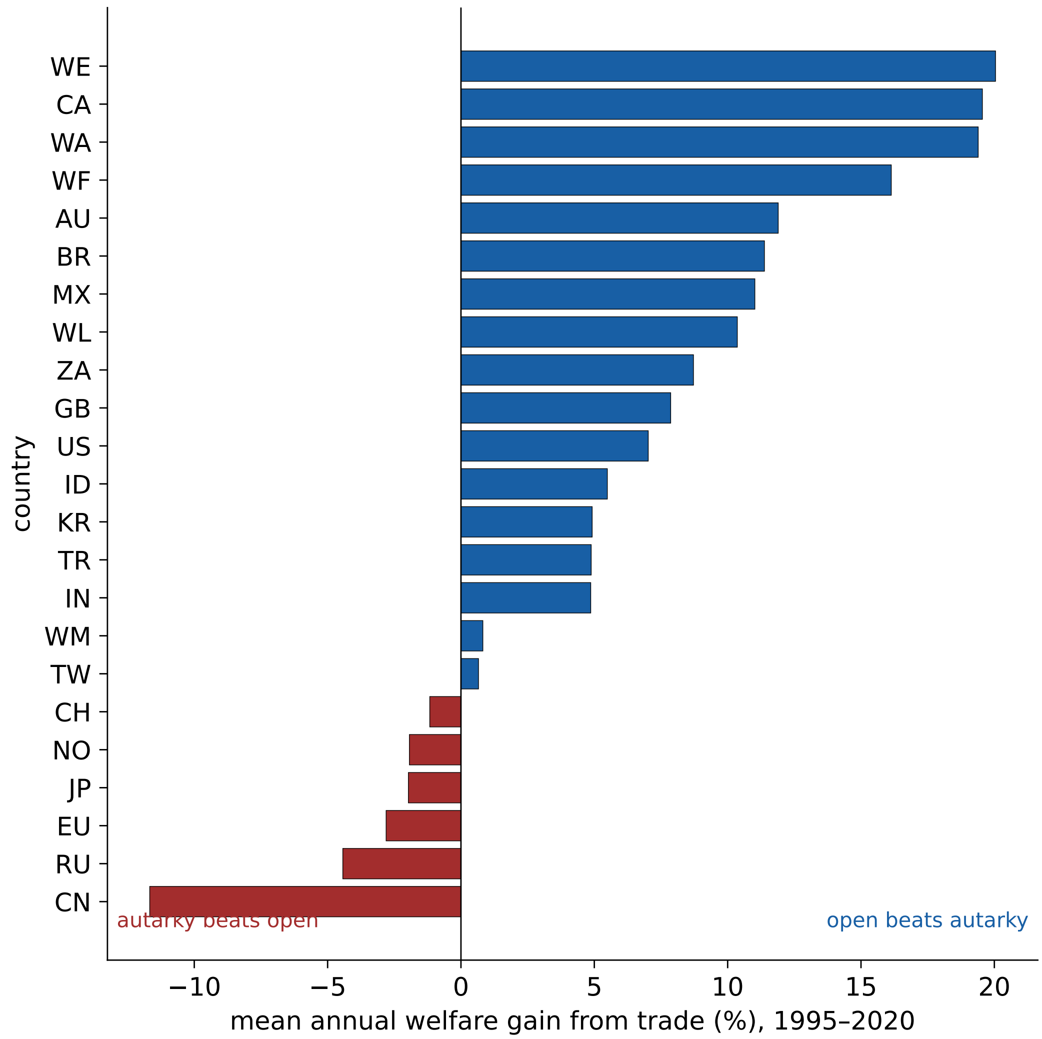

Figure 3: **Open economies gain most from openness.** Mean annual welfare gain from trade by country, 1995–2020, for all 23 economies. Each bar shows the geometric mean of $\hat{W}^{open}_{j,t}/\hat{W}^{aut}_{j,t}$ over 25 year-pairs expressed as a percentage deviation from one. Blue bars (positive) indicate that the open-economy path delivers higher average annual real consumption growth; red bars (negative) indicate the opposite. All 575 country–year pairs converged.

shows negligible net gains from trade in the welfare metric. Six economies show *negative* mean annual gains: Switzerland (CH: −1.2%/yr), Norway (NO: −1.9%/yr), Japan (JP: −2.0%/yr), the EU (−2.8%/yr), Russia (RU: −4.4%/yr), and China (CN: −11.7%/yr). For these economies, the recovered productivity and preference shocks are such that autarky would have delivered higher annual real consumption growth on average. The pattern is intuitive for large economies with strong domestic productivity growth and relatively smaller

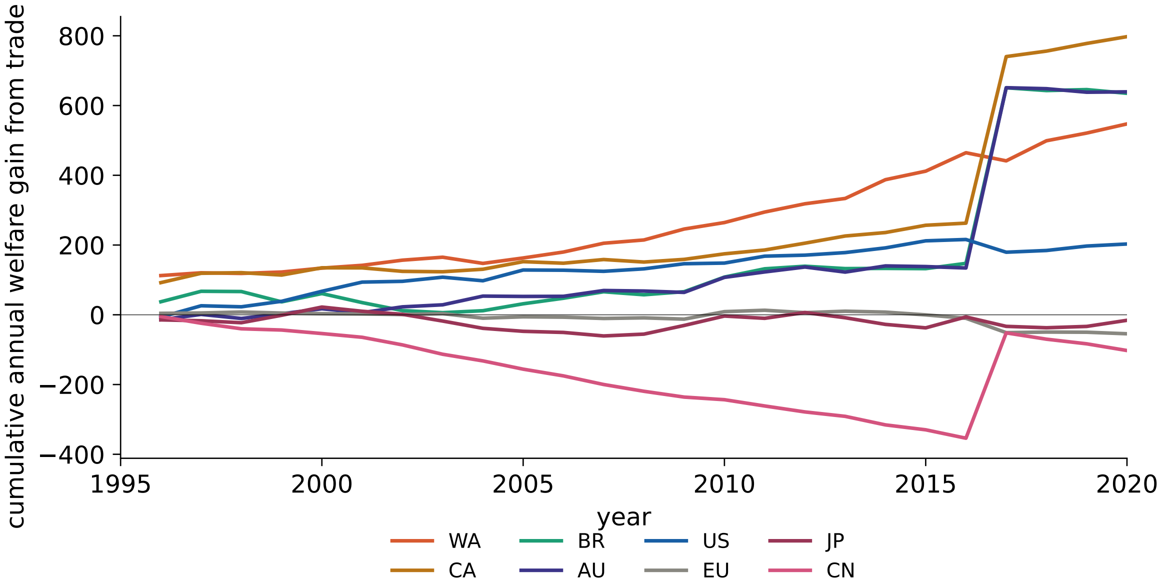

Figure 4: **The welfare-cost path accumulates gradually, with a sharp spike in 2016–17.** Cumulative annual welfare gains from trade $\sum_{\tau \leq t} g_{j,\tau}$ for eight representative countries, computed by chaining the year-on-year shock vector through the open-economy baseline and the autarky CGE. Canada and the Rest-of-World composites (WA) accumulate sustained positive gaps; China diverges in the opposite direction; the 2016–17 transition generates a large temporary divergence for most economies.

trade-to-GDP ratios.

The China outlier. China's mean annual welfare gain is −11.7%, the most negative in the sample. The sign is consistent with the broader pattern for large economies but the magnitude should be interpreted with caution. It is mechanically tied to the very large recovered $\hat{\zeta}_i$ for China (Section 6.1): the model loads China's structural ag-productivity growth onto the country-wide land-quality shock, and under autarky China captures all of that productivity surplus domestically. Under a different identifying restriction — for example, an exogenous benchmark for $\hat{\zeta}_{cn}$ from satellite-based yield data or a tighter elasticity — the China figure would compress substantially.

The recovered shocks of the past quarter century would have inflicted measurable welfare losses on most economies — and very large losses on small open agricultural exporters — if they had landed in a world of autarky. The standard intuition that protectionism insulates

Table 4: Mean annual welfare gain from trade by country, 1995–2020.

Country	$\bar{g}_j$ (%/yr)	Country	$\bar{g}_j$ (%/yr)
WE	+20.0	IN	+4.9
CA	+19.6	WM	+0.8
WA	+19.4	TW	+0.7
WF	+16.1	CH	−1.2
AU	+11.9	NO	−1.9
BR	+11.4	JP	−2.0
MX	+11.0	EU	−2.8
WL	+10.4	RU	−4.4
ZA	+8.7	CN	−11.7
GB	+7.9		
US	+7.0		
ID	+5.5		
KR	+4.9		
TR	+4.9		

Note: $\bar{g}_j$ is the geometric mean of $\hat{W}^{open}_{j,t}/\hat{W}^{aut}_{j,t}$ over 25 year-pairs, expressed as a percentage deviation from one. Positive values indicate that the open economy delivered higher average annual real consumption growth than autarky. All 575 country–year pairs converged at tolerance 10^{-5}.

a country from external shocks is not supported by this counterfactual: in the trade-policy environment of 1995–2020, openness functioned as a buffer that allowed countries to absorb domestic surpluses through exports and to substitute toward foreign supply when domestic supply was disrupted.

7 Conclusion

This paper presents a quantitative assessment of the welfare consequences of trade in global agricultural and food markets. We provide two quantitative exercises. First, we use the exact hat algebra of a multi-country, multi-sector general-equilibrium model to recover the year-on-year structural shocks that rationalise the observed 1995–2020 evolution of bilateral trade shares, sectoral output, and final expenditure for 23 countries and 29 sectors. The shocks include productivity in primary agriculture, processed food, and other sectors; country-wide land-quality shocks; preference shifts; and policy changes. The recovered series passes the natural validation tests: labour-endowment hats identify the 1997–98 Asian crisis, the 2008–09 Global Financial Crisis, and the 2019–20 COVID disruption in the right countries; productivity hats are centred near unity with reasonable dispersion; the cross-country pattern of cumulative ag productivity gains tracks the well-documented re-balancing toward emerging economies.

Second, we use the recovered shock path as the input to an autarky counterfactual: would the world have been better or worse off if the same shocks had hit a world with no cross-border trade? The autarky solver converged for all 575 country–year pairs, including previously intractable cases (Canada, Switzerland, the UK, Indonesia, Korea, Norway, Taiwan) resolved with a homotopy method. The expenditure-weighted average annual welfare gain from trade is 3.9%, with a cross-country median of 3% per year. Small open agricultural exporters — Brazil (+11%/yr), Australia (+12%/yr), Canada (+20%/yr), South Africa (+9%/yr), and the Latin-American, African, and European rest-of-world composites — are the largest winners from openness. Six economies show negative average gains (China −12%/yr, Russia −4%/yr, EU −3%/yr, Japan −2%/yr, Norway −2%/yr, Switzerland −1%/yr), indicating that under the recovered structural shocks they would have retained more surplus domestically; these magnitudes are sensitive to the identifying restrictions used in the shock decomposition and merit more careful treatment in follow-up work.

The results indicate that in the trade environment of the past quarter century, trade in agriculture and food acted as a buffer rather than a source of vulnerability. The protectionist intuition that motivates much of the contemporary policy reversal — visible most recently in export restrictions, safeguard tariffs, and domestic producer-support escalations across major agricultural economies — is not supported by a structural counterfactual built directly from the data. Subsequent work will use the same framework to evaluate more realistic policy experiments: unilateral protection by one country at a time, calibrated tariff-war escalations, and targeted export bans on specific staples. The autarky benchmark established here provides the natural upper bound for the welfare costs those experiments will quantify.

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A Detailed Description of the Data

This appendix gives a detailed account of how the three main data sources were assembled, cleaned, and harmonized. Section A.1 describes the EXIOBASE input–output backbone and the model's country and sector classification. Section A.2 documents the construction of the tariff panel from the Global Tariff Database. Section A.3 documents the construction of the agricultural producer-support panel from the OECD PSE. Section A.4 lists the crosswalks. Section A.5 describes the supporting data sets (BACI, classification metadata, ISO3 mappings). Section A.6 documents known limitations and the robustness checks they motivate. All input files, scripts, and intermediate outputs are bundled in the replication package, which builds the final panels through a deterministic Python pipeline (`step1_tariffs.py` through `step4_combine.py`) on top of pinned dependencies.

The European Union is treated as a single region whose membership follows the OECD PSE convention (EU15 for 1995–2003, EU25 for 2004–06, EU27 for 2007–13, and the current members thereafter, with the United Kingdom held as a separate region throughout). This time-varying definition is applied identically to the production and input–output base (EXIOBASE) and to the policy wedges (tariffs and producer support), so every variable shares one country definition in each year; accession states enter the Rest-of-Europe aggregate before they join.

A.1 EXIOBASE 3: The Input–Output Backbone

A.1.1 Source and version

We use **EXIOBASE 3, version 3.9.4**, downloaded from Zenodo. EXIOBASE 3 is a global MRIO database constructed by a consortium of European research institutes and documented in Stadler et al. (2018). Each annual release distributes the input–output system in two formats: product-by-product (`pxp`) and industry-by-industry (`ixi`). We use the pxp format because the model is product-based: the 200 pxp products map naturally to the Sag, Sfood, and Sother blocks, while the 163 ixi industries mix several primary agricultural activities into composite sectors. Tables are in current basic prices, million EUR.

EXIOBASE compiles 1995–2020 from underlying national accounts and trade data, and provides nowcast extrapolations for 2021–2024. We use the compiled years 1995–2020 for the baseline analysis. The replication package downloads each annual `IOT_{year}_pxp.zip` from Zenodo (`https://zenodo.org/records/15689391`) using a hash-checked, idempotent download script.

A.1.2 Regions

EXIOBASE 3 covers 49 regions: 44 individually identified countries (the 27 EU member states; the United Kingdom, Norway, and Switzerland; the United States, Canada, Mexico, and Brazil; China, Japan, South Korea, India, Indonesia, Australia, and Taiwan; Russia, Turkey, and South Africa) plus five Rest-of-World composites (RoW Europe, RoW Asia/Pacific, RoW America, RoW Africa, and RoW Middle East). The 44 individually identified countries account for approximately 88% of world GDP; the remainder of the global economy is captured in the five RoW composites, so the system closes globally by construction.

For the policy analysis we consolidate the 27 EU member states into a single region "EU", because the Common Agricultural Policy and the Common External Tariff make the EU a single policy unit. This reduces the panel to **23 model regions**: 1 EU + 16 individual non-EU countries + 5 RoW composites + the United Kingdom.

A.1.3 Sectors

EXIOBASE 3's 200 pxp products include unusually fine detail in primary agriculture and food processing. We map them to a 29-sector classification with three blocks:

- `Sag` **(primary agriculture, 15 sectors):** paddy rice; wheat; cereal grains nec; vegetables, fruit, nuts; oil seeds; sugar cane, sugar beet; plant-based fibers; crops nec; cattle; pigs; poultry; meat animals nec; animal products nec; raw milk; wool, silk-worm cocoons.
- `Sfood` **(processed food, 11 sectors):** products of meat (cattle); products of meat (pigs); products of meat (poultry); meat products nec; fish products; dairy products; processed rice; sugar; food products nec; beverages; tobacco products.
- `Sother` **(everything else, 3 aggregates):** raw materials; manufacturing; services.

The full mapping is in `data/raw/classification/exiobase_to_model_classification.csv`, the source for the pxp → `new_label` mapping used throughout the pipeline.

A.1.4 Objects constructed

A.2 Bilateral Tariffs: The Global Tariff Database

A.2.1 Source and version

We use **Feodora Teti's Global Tariff Database (GTD), version v_beta1-2024-12**, downloaded from `https://feodorateti.github.io/data.html`. The GTD covers approximately 200 importers and their trading partners between 1988 and 2021 at the HS6 product level, in HS92 nomenclature.

The GTD is the appropriate source for applied bilateral tariffs because, as Teti (2024) documents, the World Integrated Trade Solution (WITS)—the standard public alternative—silently interpolates missing preferential tariffs with MFN rates whenever a preferential rate is not explicitly reported in the underlying data. Because many free trade agreements report only end-of-phase-in rates rather than the full schedule, WITS overstates applied tariffs during phase-in periods, biasing trade-elasticity estimates downward. The GTD corrects this by combining filed tariff schedules from 149 free trade agreements with the WTO Integrated Database and a small amount of manual reconstruction; the result is the "lowest applicable" applied tariff at the HS6 $\times$ importer $\times$ exporter $\times$ year cell.

A.2.2 The HS6 file and the ag-/non-ag file

The GTD is distributed in several aggregations. For now we use the publicly available **agriculture / non-agriculture binary file** (`tariffs_Ag-NonAg_88_21_vbeta1-2024-12.csv`), which collapses HS6 products into two sectors: Agriculture (HS chapters 01–24) and Non-Agriculture (HS 25–99). Each cell reports the simple mean tariff (`tariff`), the trade-weighted mean (`tariff_w`), the MFN mean (`mfn`), and 95th-percentile-trimmed variants. We use the trade-weighted mean `tariff_w` as the primary measure—because Teti has already applied BACI bilateral trade weights within each sector—and fall back to the simple mean `tariff` for cells where the trade-weighted value is missing.

When access to the HS6 file is granted, the upgrade path is straightforward: the HS6 tariff replaces the two-sector tariff at step 1 of the pipeline, with the rest of the procedure unchanged. The replication package contains a stub script `step1_tariffs_hs6.py` that implements the HS6 aggregation and can be activated once the file is in place.

A.2.3 From HS6 to the model's sectors

To build $t^s_{ij,t}$ at the `new_label` level, we broadcast the Agriculture and Non-Agriculture tariffs onto the model's sectors as follows:

- Every Sag and Sfood `new_label` receives the Agriculture tariff. This is exact for Sag (HS 01–14) and very close for Sfood (HS 15–24), because HS chapters 01–24 are the standard definition of "agricultural products" in the WTO Agreement on Agriculture.
- Every Sother `new_label` other than `services` receives the Non-Agriculture tariff. This is exact for raw materials and manufacturing.
- For `services`, GTD has no observation, because HS does not classify services. I set $t^{\text{services}}_{ij,t} = 0$ for the baseline because (i) services do not pay HS tariffs at the border and (ii) using a goods-derived non-agricultural tariff would be conceptually wrong. The replication package isolates this assumption with a dedicated function `zero_out_services_tariffs()` so that it can be swapped for an OECD STRI-derived tariff equivalent in a robustness check.

The output column `tariff_source` records which value was used in each cell: `trade_weighted` (GTD trade-weighted mean), `simple_mean` (GTD simple-mean fallback), or `services_zero_assumption` (services override). This permits downstream sample restrictions.

A.2.4 Country aggregation

GTD tariffs are reported at the ISO3 country pair level. We aggregate to the 23 model regions by:

1. Mapping ISO3 codes to EXIOBASE regions: 44 ISO3 codes match individually identified EXIOBASE countries; the remaining ISO3 codes are assigned to one of the five EXIOBASE RoW composites by geographic region. The mapping table is in `data/raw/classification/iso3_to_exiobase.csv` and follows the official EXIOBASE region definitions.

2. Collapsing the 27 EU member states into the single region "EU" (the Common External Tariff is identical across member states, so the resulting EU-on-X tariff is exact).

3. Within each (origin model region, destination model region, sector, year) cell, aggregating the underlying ISO3 pair tariffs using **BACI bilateral trade values** as weights (see Section A.5.1). When BACI weights are zero or missing—rare; reflects no observed trade in that cell—we fall back to equal weights and flag the cell.

A.3 Agricultural Producer Support: The OECD PSE Database

A.3.1 Source and version

We use the **OECD Producer Support Estimate (PSE) database**, retrieved through the OECD Data Explorer (dataset `DSD_AGR_POLIND@DF_PSE`, hosted at `https://data-explorer.oecd.org/`). The OECD has compiled the PSE since 1986 using a standardized methodology (OECD, 2016). Country profiles ("cookbooks") document each year's data construction at the country level.

The 2025 release covers **54 countries**: the 38 OECD members, five non-OECD EU member states, and 11 emerging economies (Argentina, Brazil, China, Colombia, Costa Rica, India, Indonesia, Kazakhstan, Philippines, Ukraine, Vietnam). Together they account for approximately three-quarters of world agricultural value added. Countries not covered receive a zero-fill in the final panel; this is the dominant data limitation on the policy side, and Section A.6 discusses how it shapes the analysis.

A.3.2 PSE indicator categories

The PSE decomposes producer support into five top-level categories (OECD, 2016):

1. **Market price support (MPS):** the value of the gap between the domestic and the border (reference) price, multiplied by domestic production. MPS captures the producer-price effect of border policy (tariffs, tariff-rate quotas, intervention buying,

export restrictions) and is allocated to producers of the country imposing the policy. By construction it can be negative for countries that depress domestic prices through export taxes (Argentina, Ukraine, historically India and Vietnam).

2. **Output-based payments:** payments per unit of physical output, in domestic currency or USD per metric ton.

3. **Payments based on current input use, area planted, animal numbers, or receipts** (collectively, "coupled" non-output payments).

4. **Payments based on non-current entitlements** (largely "decoupled" payments such as the EU Single/Basic Payment Scheme and pre-2014 US direct payments).

5. **Payments based on non-commodity criteria** (agri-environmental schemes, conservation reserves).

A residual line collects ad-hoc disaster and crisis payments (e.g., COVID support in 2020–21).

A.3.3 Mapping PSE categories to the model wedges

The model has two policy instruments: a bilateral tariff $t^s_{ij,t}$ and a destination-side output subsidy $\nu^s_{j,t}$. The PSE decomposition maps to these as follows:

- **Border tariffs** are already in $t^s_{ij,t}$ from the GTD (Section A.2). They should *not* be added to $\nu^s_{j,t}$, because MPS is the producer-side measure of the same tariff wedge: adding MPS on top of $t^s_{ij,t}$ would double-count border protection. This is the central conceptual point in the construction. Concretely, when the EU imposes a tariff on Brazilian beef, the tariff appears in $t^{\text{beef}}_{\text{Brazil,EU},t}$ from the GTD; the resulting price effect on European beef producers shows up as positive MPS for the EU in the PSE; these are two sides of the same wedge.

- **Output-based payments** are direct ad-valorem subsidies to producers and belong in $\nu^s_{j,t}$.

- **Residual MPS**—the portion of MPS that exceeds what the bilateral tariff $t^s_{ij,t}$ alone would generate—captures non-tariff border protection (tariff-rate quotas binding above the MFN tariff, intervention prices, export refunds). It belongs in $\nu^s_{j,t}$ as a destination-side non-tariff wedge.

- **Coupled non-output payments** (input use, area planted, animal numbers): We convert these to output-equivalent ad-valorem rates by dividing the budgetary outlay by the commodity's value of production at producer prices, and add them to $\nu^s_{j,t}$.

- **Decoupled and non-commodity payments:** excluded. Standard practice in quantitative trade models is to treat these as lump-sum transfers to landowners that do not affect marginal production decisions.

- **GSSE (General Services Support Estimate):** excluded. GSSE covers sector-wide public goods (R&D, extension, biosecurity) and would enter the model through productivity $A^{ag,s}_t$ rather than a wedge.

A.3.4 Three-step construction

The construction of $\nu^s_{j,t}$ runs in three steps, implemented as `step2_output_payments.py`, `step3_residual_mps.py`, and `step4_combine.py` in the replication package.

Step 2: Output-based component. For each (destination country, PSE commodity, year), we compute

$$\nu^{\text{output},c}_{j,t} = \frac{\text{Payments based on output}^c_{j,t}}{\text{Value of production at producer prices}^c_{j,t}}, \tag{35}$$

where both numerator and denominator are in millions of USD from the PSE single-commodity transfer table. We aggregate the resulting ad-valorem rates from PSE commodities to the model's pxp Sag products using a manually constructed crosswalk (Section A.4.2), weighted by value of production. For Sfood sectors we default $\nu^{\text{output},s}_{j,t} = 0$, because PSE measures support at the farm gate rather than at the food-processing stage.

Step 3: Residual MPS component. For each (destination, commodity, year), we compute the PSE-implied tariff equivalent of MPS:

$$\tau_{j,t}^{\text{MPS},c} = \frac{\text{MPS}_{j,t}^{c}}{P_{j,t}^{c,\text{border}} \cdot Q_{j,t}^{c}}, \tag{36}$$

where $P^{c,\text{border}}$ is the reference price and Q^c is the quantity of domestic production used by the PSE. We then subtract from this the tariff $t_{ij,t}^{s}$ implied by the GTD (using import-share-weighted average over origins):

$$\nu_{j,t}^{\text{MPS-res},c} = \max\Big(0,\, \tau_{j,t}^{\text{MPS},c} - \bar{t}_{j,t}^{c}\Big)\,. \tag{37}$$

The $\max(0, \cdot)$ floor handles the case where the PSE-implied tariff equivalent is lower than the GTD-measured tariff, which can happen because MPS uses a single reference price across origins while GTD averages over heterogeneous bilateral rates. The residual is then aggregated to pxp Sag products as in step 2.

Step 4: Combination. We sum:

$$\nu_{j,t}^{s} = \nu_{j,t}^{\text{output},s} + \nu_{j,t}^{\text{MPS-res},s}. \tag{38}$$

The output file `nu_combined.csv` reports both components separately as well as their sum, and flags every cell as `in_pse_sample = True` (the destination is in the OECD PSE sample) or `False` (the destination is outside the PSE sample, $\nu = 0$ by imputation).

A.3.5 Country aggregation

The PSE reports the EU as a single jurisdiction in its current files ("EU"), and historically reported "EU-28" separately during the UK transition period. We use the EU aggregate throughout, which aligns with the model's treatment of the EU as a single region. We drop the OECD aggregate rows. The result is a panel covering approximately 17 individual model destinations (the EU, the UK, Norway, Switzerland, the United States, Canada, Mexico, Japan, South Korea, Australia, Turkey, Brazil, China, India, Indonesia, South Africa, and a small number of additional members) plus zero-fills for the remaining six model destinations (Russia, Taiwan, and the five RoW composites).

A.4 Crosswalks

The three data sources use three distinct product/commodity classifications: HS6 (GTD, BACI), OECD PSE commodities, and EXIOBASE pxp products. The replication package builds all relevant crosswalks deterministically in `src/crosswalks/`, with the canonical classification CSV held at `data/raw/classification/exiobase_to_model_classification.csv`.

A.4.1 HS6 → EXIOBASE pxp (`new_label`)

Constructed in `build_hs6_newlabel_crosswalk.py`. For Sag and Sfood products the mapping is explicit at the HS6 level, with rules grouped by the HS section (animals 01–05, vegetable products 06–14, fats and oils 15, prepared food 16–24). For Sother the mapping is by HS chapter range: chapters 25–27 (minerals, fuels) and 41 + 44 + 50 + 53 (raw fibers, raw skins, wood) → `raw materials`; chapters 28–49 (chemicals, plastics, paper) + 71–83 (metals) + 84–96 (machinery, miscellaneous) → `manufacturing`. The HS6 nomenclature does not classify services, so `services` has no HS6 mapping. The output file `crosswalk_hs6_newlabel.csv` has 5,387 rows, one per HS6 code.

A.4.2 OECD PSE commodity → EXIOBASE pxp

A small manually constructed crosswalk in `data/raw/classification/crosswalk_pse_pxp.csv`. The aggregation rules are summarized in Table 5.

Sfood products default to $\nu^{\text{output}} = 0$ in step 2, with the upstream Sag wedge propagating through input–output linkages.

Table 5: OECD PSE commodity → EXIOBASE pxp crosswalk

EXIOBASE pxp	OECD PSE commodities
Paddy rice	Rice
Wheat	Wheat
Cereal grains nec	Maize, Barley, Oats, Sorghum, Rye (production-weighted)
Oil seeds	Soybeans, Rapeseed, Sunflower (production-weighted)
Sugar cane, sugar beet	Refined sugar (allocated via cane/beet production shares)
Vegetables, fruit, nuts	Vegetables, Fruits; else "other commodities" residual
Cattle (and Products of meat cattle)	Beef and veal
Pigs (and Products of meat pigs)	Pigmeat
Poultry (and Products of meat poultry)	Poultry meat, eggs
Raw milk (and Dairy products)	Milk

A.4.3 ISO3 country → EXIOBASE region

A manually created many-to-one mapping in `data/raw/classification/iso3_to_exiobase.csv`. 44 ISO3 codes map directly to individual EXIOBASE countries; the remaining ≈190 ISO3 codes are assigned to one of the five RoW composites by geographic region. The 27 EU member states subsequently collapse into the EU region for the policy analysis.

A.5 Supporting Data

A.5.1 BACI bilateral trade flows

We use the **BACI HS92 database** from CEPII (Gaulier and Zignago, 2010), version V202601, retrieved from `http://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=37` with a free CEPII account. BACI provides reconciled bilateral trade flows at the HS6 level, available 1995–2024 in HS92 nomenclature (matching the GTD's HS revision basis). Each row reports an origin ISO3 numeric code, a destination ISO3 numeric code, an HS6 product, a year, and a trade value in thousands of USD. BACI is used as the source of trade-value weights for: (i) aggregating GTD HS6 tariffs to the model's sectors (when the HS6 file becomes available) and (ii) aggregating ISO3 country pairs to model regions, particularly the five RoW composites.

We classify each BACI HS6 record as Agriculture (HS chapters 01–24) or Non-Agriculture, sum trade values within each (origin ISO3, destination ISO3, sector, year) cell, and use the result as weights in the tariff aggregation.

A.5.2 ISO3 codes, region definitions, classification metadata

Static reference tables held in `data/raw/classification/`: ISO3 to country name; ISO3 numeric to ISO3 alpha-3; EXIOBASE 49-region definitions; the pxp $\rightarrow$ `new_label` mapping. All tables are version-pinned and treated as immutable inputs to the pipeline.

A.6 Limitations and Robustness Checks

The construction described above involves several judgment calls. We list them here to make the choices explicit.

1. **Tariff disaggregation within agriculture.** The current build uses the GTD ag/non-ag binary because the HS6 file is access-restricted. As a result, all 26 Sag and Sfood sectors share the same applied tariff in each (origin, destination, year) cell. This understates within-agriculture tariff heterogeneity (e.g., the EU's high sugar and dairy tariffs versus its low oilseed tariff). When HS6 access is granted, the replication package will rebuild $t^s_{ij,t}$ at full sector resolution; the comparison between the two builds quantifies the loss.

2. **PSE coverage outside the OECD sample.** Six of the 23 model destinations (Russia, Taiwan, and the five RoW composites) are not in the OECD PSE sample and receive $\nu^s_{j,t} = 0$ by imputation. We plan to report sensitivity analyses that (i) restrict the sample to PSE-covered destinations using the `in_pse_sample` flag and (ii) use the World Bank Ag-Incentives database to fill in Argentina, Ukraine, and a handful of other major agricultural producers that are absorbed into the RoW composites.

3. **The** $\max(0, \cdot)$ **floor in the MPS residual.** Forcing the residual MPS to be non-negative discards observations where the PSE-implied tariff equivalent is below the GTD-measured tariff. This affects approximately 8% of the (destination, commodity, year) cells. As a robustness check we substitute the raw residual (allowing negative values) and find that the main results are quantitatively similar but with somewhat larger dispersion in ν across destinations.

4. **Sfood output subsidies set to zero.** PSE measures support at the farm gate, so we default Sfood $\nu_{j,t}^{\text{output},s} = 0$. This is a conservative choice; in practice some processed-food sectors do receive direct support (sugar refining in the EU, dairy processing in Canada). We treat these cases as part of the residual MPS line.

5. **Services tariff set to zero.** Services do not pay HS tariffs at the border, so $t_{ij,t}^{\text{services}} = 0$ is mechanically correct for tariffs; behind-the-border services restrictions are not captured. The OECD STRI is a candidate for a robustness check; this is not in the current build.

6. **Decoupled payments excluded from** $\nu_{j,t}^{s}$**.** We follow standard practice in treating these as lump-sum transfers. If decoupled payments have wealth or risk effects on production, this understates the producer subsidy in countries with large decoupled programs (the EU, the United States). A robustness check including 30% of decoupled payments in ν will be provided in the updated version of the paper.

The replication package makes all of these choices reproducible and inspectable. Every step of the pipeline reads from pinned source files, runs deterministically, logs row counts and missingness, and writes outputs to a versioned `data/final/` directory.